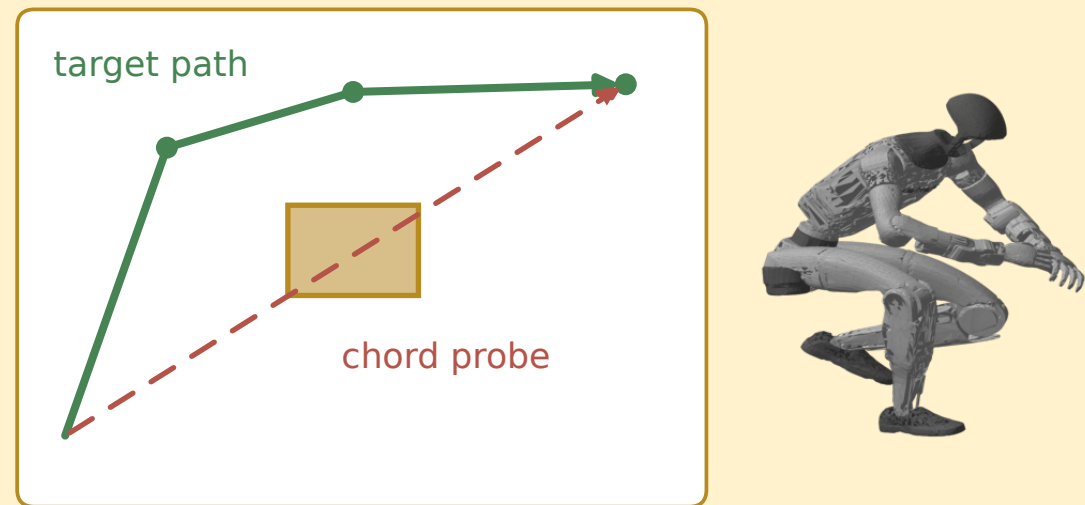


Motion2Scene: from motion evidence to scene-and-goal control

Three linked learning problems: infer useful geometry, preserve movement capability, and generate commands from task context.

(a) Learn scenes from motion

Full motion → event-conditioned proposer



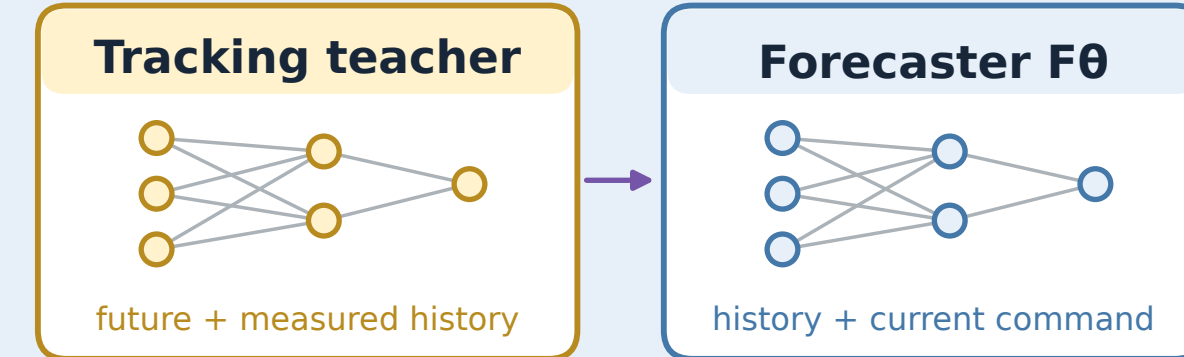
Hindsight: preserve the observed movement; place geometry that challenges an alternative.

225 fixed recipes → learned categorical scores

Clearance mask → sequential scene sampling

Schematic geometry; reference pose, not a rollout.

(b) Recover the full-command motor



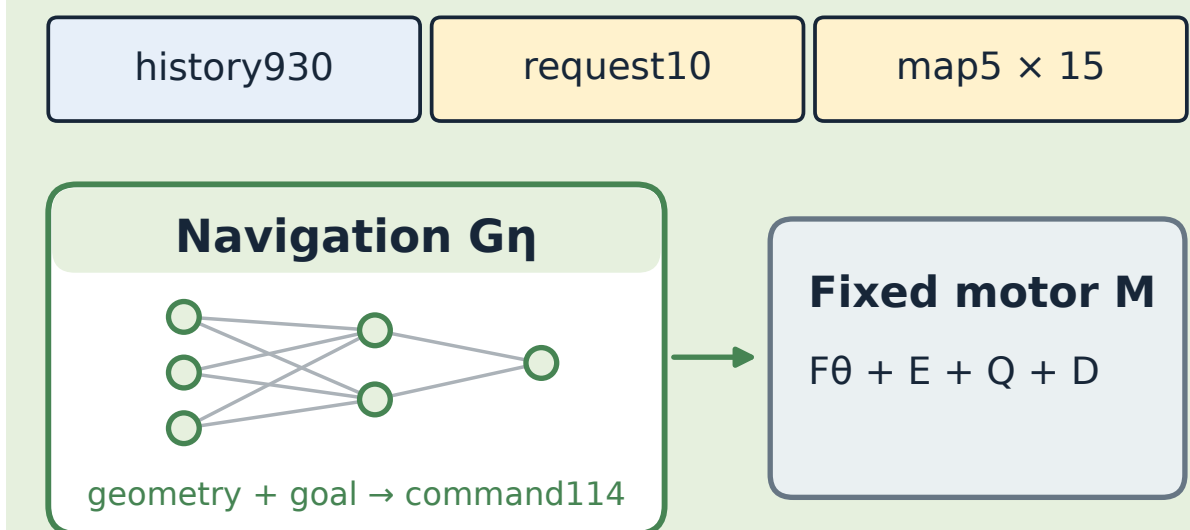
Online action / token distillation

Predicted reference → fixed E → Q → D

Retained motor representation → 29D action

Teacher recovery and scene fitting are separate studies.

(c) Learn navigation from context



Full command is completed internally.

Approach → brake → stable goal hold

Known-map control; no reference at inference

Camera context and wider tasks remain research branches.

(d) Convert compatible scenes and motor capability into qualified supervision

Scene + goal → execute continuation → score task → admit supported rows

Fresh 50-tick hold; distance, speed, contact, fall and deadline. Retain failed attempts and collection cost.

Learner-state recovery supplies bounded local support; scene acceptance alone supplies no action label.

Fit navigation; roll out again

Current evidence: 4/8 initial tasks,
1/8 on evaluation-seed confirmation.
End-to-end data utility remains unproven.